Spaceship Earth – from attractive metaphor to architectural analogue

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Grateful acknowledgements to:
Bruce McNaughton, Luke Moffat and Lewis Bowick, INCOSE UK Energy Systems Interest Group
Morris Bradley and other members of the Edinburgh U3A Climate Change Group

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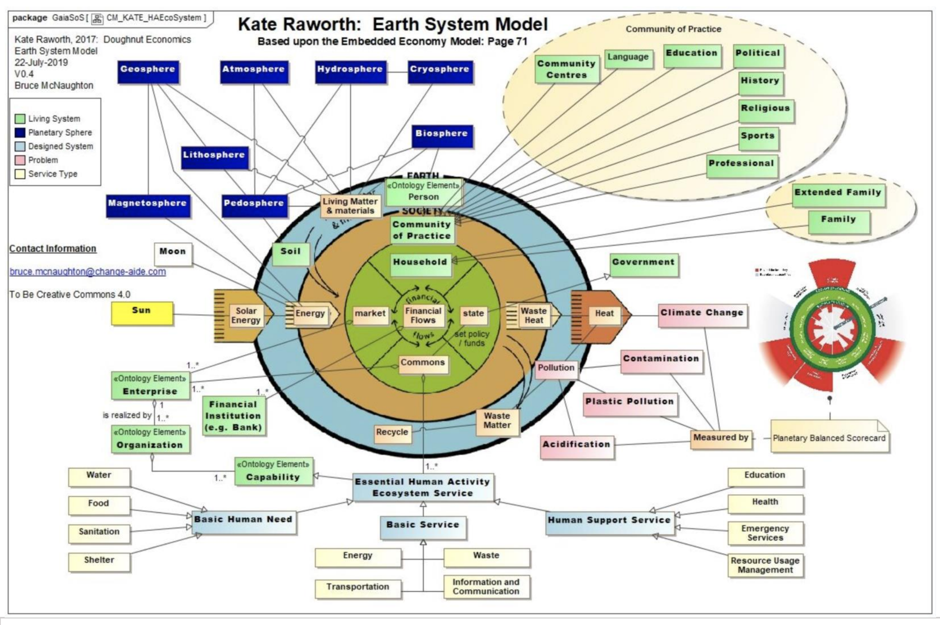


The Economics of the Coming Spaceship Earth

Kenneth E. Boulding

In H. Jarrett (ed.) 1966. Environmental Quality in a Growing Economy, pp. 3-14. Baltimore, MD: Resources for the Future/Johns Hopkins University Press.

First presented by Kenneth Ewart Boulding at the Sixth Resources for the Future Forum on Environmental Quality in a Growing Economy in Washington, D.C. on March 8,1966.

We are now in the middle of a long process of transition in the nature of the image which man has of himself and his environment. Primitive men, and to a large extent also men of the early civilizations, imagined themselves to be living on a virtually illimitable plane. There was almost always somewhere beyond the known limits of human habitation, and over a very large part of the time that man has been on earth, there has been something like a frontier. That is, there was always some place else to go when things got too difficult, either by reason of the deterioration of the natural environment or a deterioration of the social structure in places where people happened to live. The image of the frontier is probably one of the oldest images of mankind, and it is not surprising that we find it hard to get rid of.

2

Unlimited space and resources – unlimited sink for waste - "open planet" – image of the frontier

- World is chaotic, *act-sense-respond*
- Extract, exploit, consume, move on to pastures new
- Move fast and break things!
- No limits to growth, no problems I can't deal with,
- Challenge and test boundaries
- Plan B is for wimps!
- Competition, winner takes all
- Self-actualisation through accumulating wealth and status at the expense of
  - a) others,
  - b) nature

Economics of Frontiersman era don't work in the Anthropocene –
- Focus on production/consumption/growth,
- Focus on maximising flow
- Maintenance is viewed as deferable cost
- Poor social outcomes:

- **Capitalism** leads eventually to monopoly and inequality ("success to the successful", trickle-down doesn't prevent rising inequality);
- focus on in-year profit and high discount rates lead to short termism, tragedy of the commons

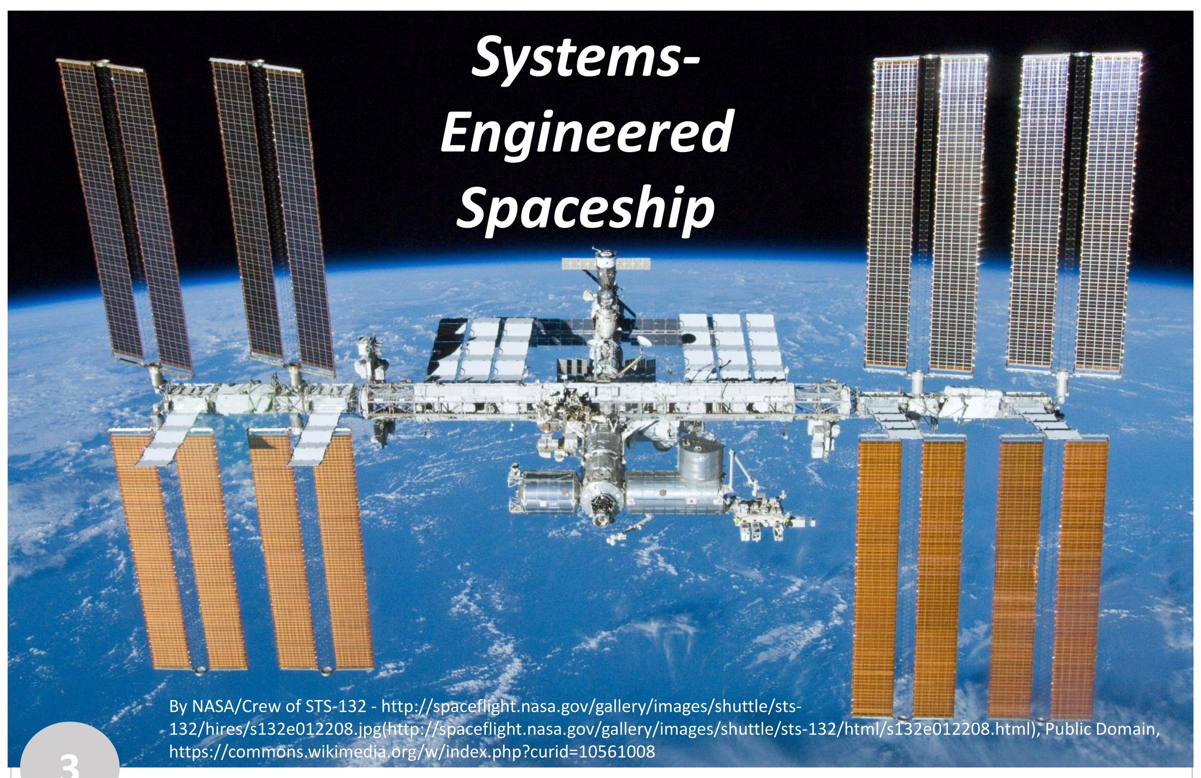
- **Communism** ignores human nature, suppresses self actualisation, allocates resources inefficiently

Limited space and resources – "closed planet" – image of the spacecraft

- World is complex, *probe-sense-respond*
- Actions may have delayed consequences
- Conserve capital, maintain dynamic equilibrium
- Take reasonable care, anticipate problems
- Understand and don't violate boundaries, safety, dynamic and integrity limits
- Have a Plan B for likely eventualities
- Collaboration, seek win-win outcomes
- Self-actualisation through mission success, caring for others

Economics of Spaceman era for the Anthropocene??
- Focus on wellbeing and harmony,
- Focus on conserving stock,
- Maintenance is vital
- Produce good social outcomes:

- A third way for economics?
  - Circular economy?
  - Doughnut economics?
- Avoid excesses of both Capitalism and Communism.
- Focus on "whole mission", Consider future generations
- Continue to offer opportunities for self actualisation, but that don't harm others,
- How do we design a spaceship for 9 Billion people?
- How do we keep a crew of 9 billion content?



By NASA/Crew of STS-132 - http://spaceflight.nasa.gov/gallery/images/shuttle/sts-132/hires/s132e012208.jpg(http://spaceflight.nasa.gov/gallery/images/shuttle/sts-132/html/s132e012208.html), Public Domain, https://commons.wikimedia.org/w/index.php?curid=10561008

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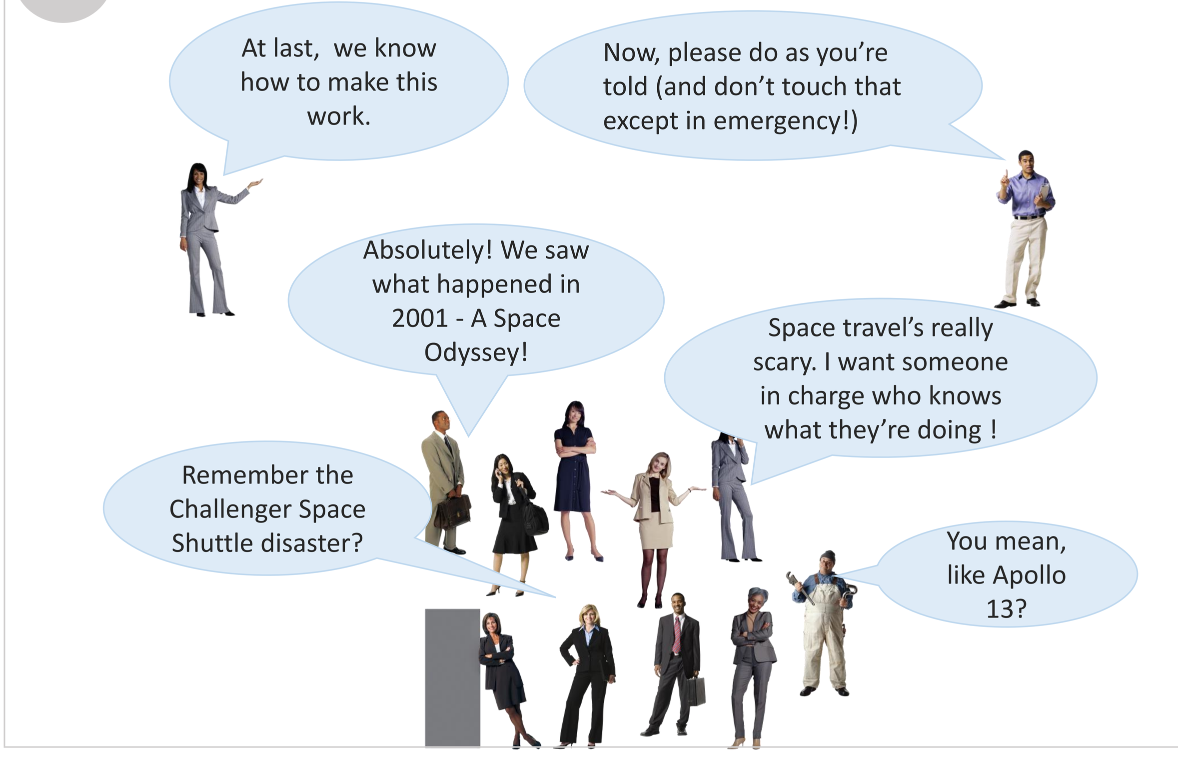

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Systems Engineering is a transdisciplinary and integrative approach to enable the successful realization, use, and retirement of engineered systems, using systems principles and concepts, and scientific, technological, and management methods.

We use the terms "engineering" and "engineered" in their widest sense: "the action of working artfully to bring something about". "Engineered systems" may be composed of any or all of people, products, services, information, processes, policies, and natural elements.

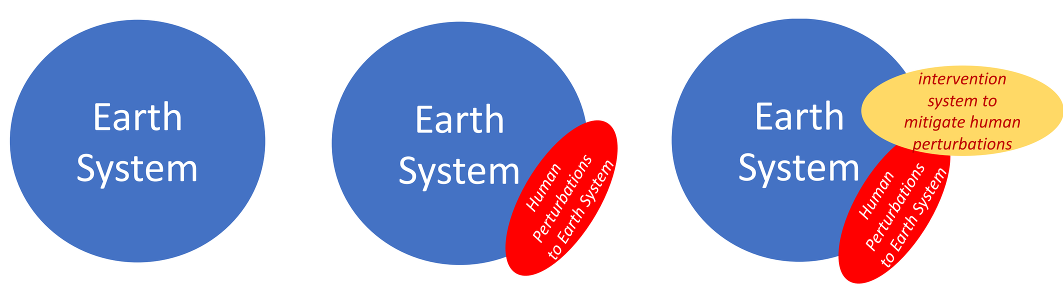
An **engineered system** is a system designed or adapted to interact with an anticipated operational environment to achieve one or more intended purposes while complying with applicable constraints.

Thus, an "engineered system" is a system – not necessarily a technological one - which has been or will be "systems engineered" for a purpose. A very general and inclusive definition of "system" is the following:

A **system** is an arrangement of parts or elements that together exhibit behaviour or meaning that the individual constituents do not.

A **physical** system exhibits **behaviour**
A **conceptual** system exhibits **meaning**

(Sillitto et al, INCOSE Fellows' Definitions report, 2019)



Human interventions designed to to restore Gaia to its prior equilibrium, or to bring it to a new equilibrium still compatible with human society, will be an "engineered system".

For this system to be successful, we will need a transdisciplinary approach to bring together and synergise the knowledge and efforts of ecologists, earth system scientists, systems engineers, system scientists, and many many many others. – i.e. transdisciplinary Systems Engineering.

ALL will have to come out of their silos, think differently and ***LEARN TOGETHER***.

A successful transdisciplinary approach requires a shared mental model (Senge).

Can an elaborated view of SPACESHIP EARTH become the required shared mental model? Should it?

Logic:
1. Few people understand Gaia – many (kind of) understand spaceships.
2. The FUNCTIONS of Gaia supporting life on Earth are broadly the same as the FUNCTIONS of a spacecraft designed for a long term unsupported mission.
3. Potential to use a common FUNCTIONAL (or LOGICAL) architecture to describe a) spaceships and b) spaceship earth.
4. The spacecraft is easier to understand because as a designed engineered system it has a relatively simple mapping of functions to physical elements.
5. Gaia is more difficult to understand because as an evolved natural system it is highly complex, its component systems are deeply enmeshed and highly coupled with no simple function-to-physical mapping.
6. We can use familiar spaceship analogues to understand the less familiar Gaia issues.
7. Transforming our thinking from the physical into the functional domain is key to establishing useful analogues between different but similar systems.

Key difference is that we have more control over the elements of a spaceship than we have over the elements of Gaia. But that's OK, we can work with that.

Performance and safety envelope, dynamic and integrity limits

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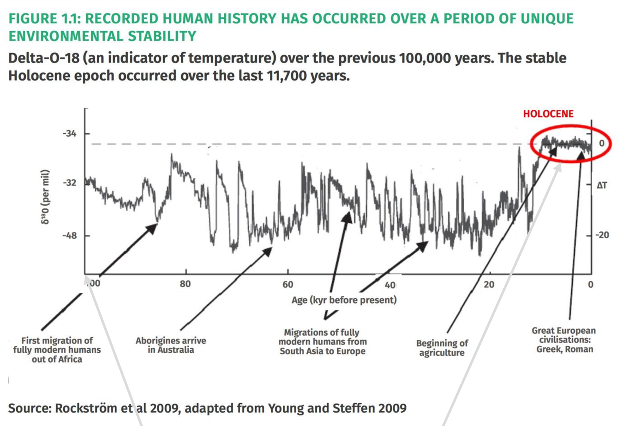
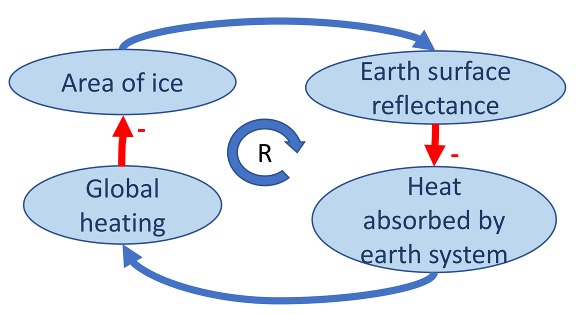
Holocene conditions are UNPRECEDENTED in human existence, and possibly in the geological record.

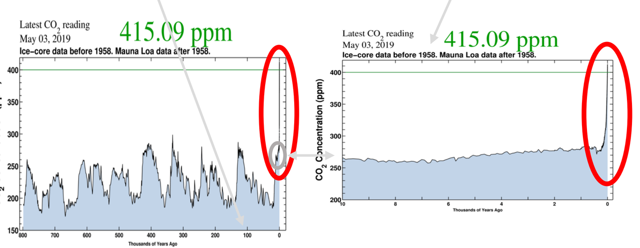
There is NO EVIDENCE that complex human civilisations can exist in conditions other than Holocene.



Source: Rockström et al 2009, adapted from Young and Steffen 2009



Human activity is increasing CO_2 in atmosphere at rate unprecedented in geological record. Current level is probably incompatible with long term survival of ice sheets.
Ice sheets are Earth's solar shield, they stop it getting too hot.



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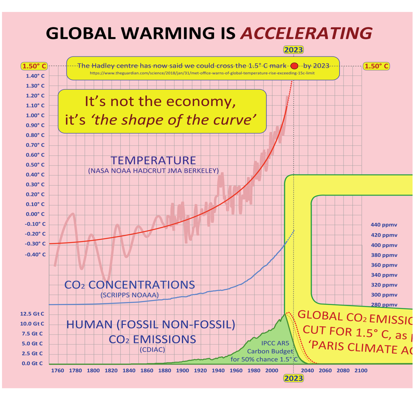
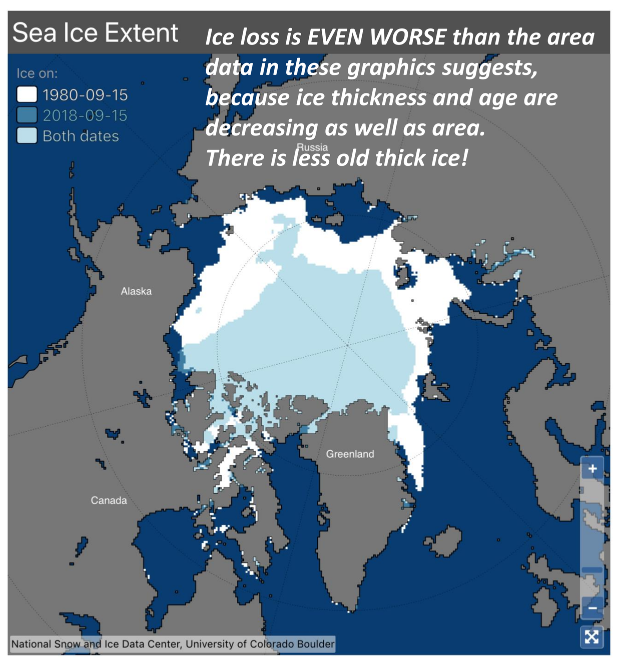
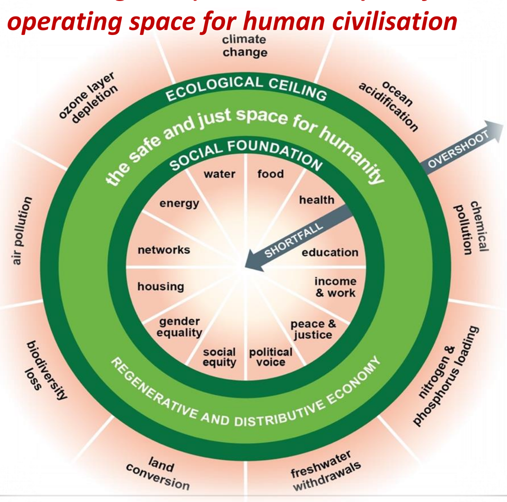
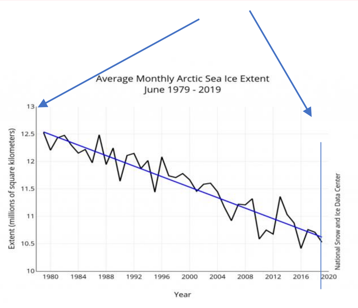




The Doughnut (Kate Raworth) – safe operating space for human civilisation





Likely to see no arctic sea ice in summer sometime between 2020 and 2050.
*"Houston, we have a problem". But there is **no ground control for Spaceship Earth** – control emerges bottom up, from us and the rest of Gaia!*

North Polar ice melt is the canary in the mine – the canary is very ill - why have we not acted on the warning yet?

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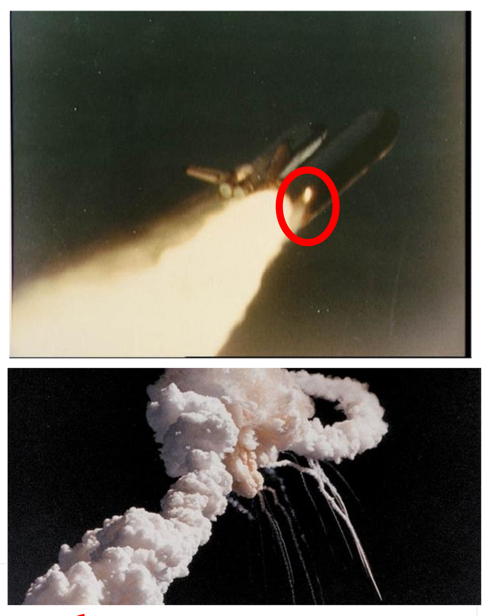
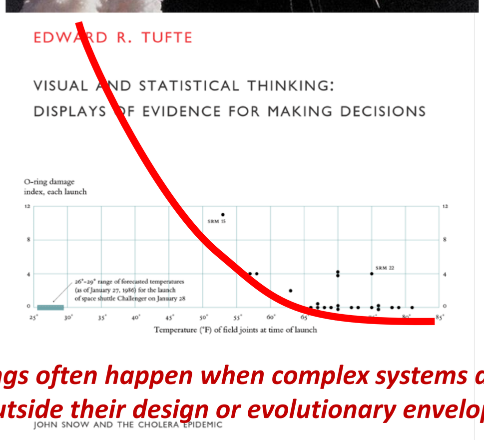




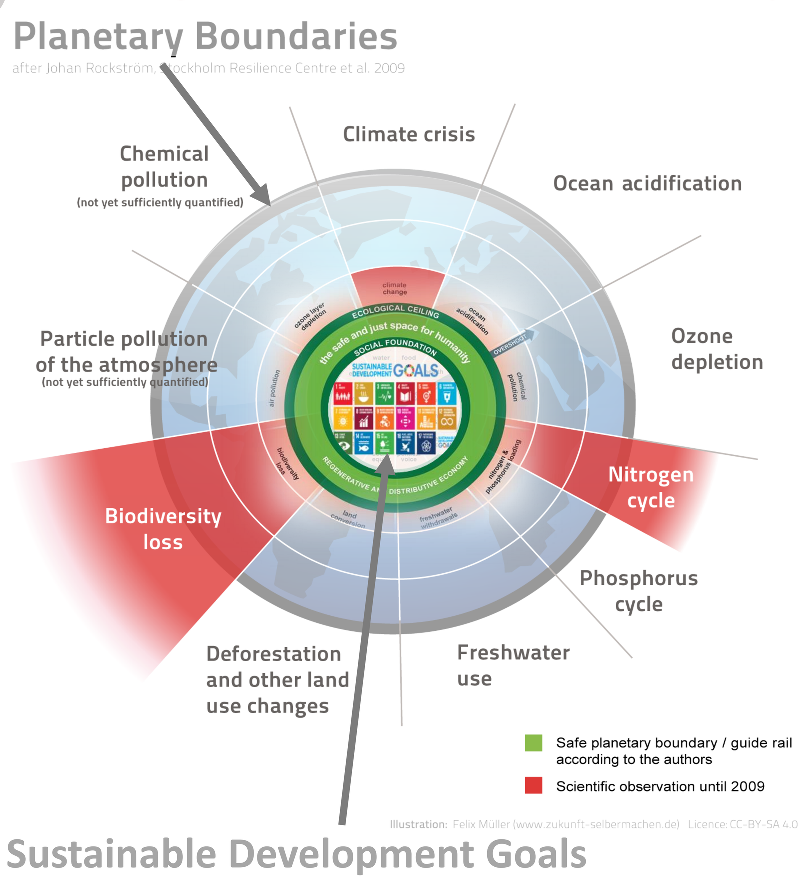
Bad things often happen when complex systems are taken outside their design or evolutionary envelope.

***Challenger Shuttle disaster** – frontier mentality led to launch in MUCH colder conditions than*
- *a) previously experienced,*
- *b) O-rings were certified to,*
- *c) O-rings stayed flexible.*

In technical engineered systems
- *a) Complexity is whenever possible encapsulated within modules and subsystems*
- *b) No guarantee of performance or safety if system is taken outside design and test limits.*
- *c) System may go HIGHLY non linear if safe operating space transgressed.*

Challenger disaster due to over confident managers over riding carefully phrased engineering advice – "we just don't know" was heard as "probably all right".

7 The Doughnut – balanced scorecard for Spaceship Earth



8 Spaceship Earth Architectural Concept

Earth system physical architecture

Common logical (= functional) architecture for Earth and Spaceship systems – network of interacting processes that we know how to map to both Earth and Spaceship physical architectures.

Spaceship system physical architecture

Earth system physical element	Function/process	Spaceship system physical element
Atmosphere, Biosphere, Gaia's Homeostasis, etc	Maintain conditions for life – including: Provide supply of air, water and food Maintain temperature Manage waste Shield the biosphere from harmful influences (radiation, micro-meteorites, etc).	Environmental Control & Life Support System (ECLSS)
Gravity	Keep atmosphere contained	Pressure hull
Photosynthesis	Capture and convert solar energy into usable/storable form	Solar panels
Ice sheets, clouds, aerosols	Reflect away excess incident solar radiation	Solar shields
etc	etc	etc
etc	etc	etc

9 Next steps:

*Move "spaceship earth" from an attractive metaphor, to a precise **and widely understood and used** formal analogue. This is so we can*
- *a) draw precise, and ideally quantitative, parallels between earth system issues and spaceship system issues,*
- *b) use 'familiar' spaceship analogies to explain the hard-to-understand Gaia systems and how to restore planetary health.*

1. Is this worth doing?

2. If so, how to do it?
- *a. Research programme?*
- *b. Open source collaboration?*
- *c. Competition with cash prizes?*
- *d. c followed by a or b?*
- *e. And then…???*

Version 1.2, 7th August 2019
Minor changes from version 1.0, which was Presented at Lovelock Centenary Conference, Exeter, 29-31 July 2019